PETRA ZLATARIĆ-NOVOSEL

Machine Learning Engineer

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PROFESSIONAL SUMMARY

Machine learning engineer with more than five years of experience in natural language processing and document classification, from research and prototyping to production. Experienced in transformer fine-tuning, model serving and monitoring on Azure, and customer complaint classification in banking. PhD candidate researching named entity recognition in Croatian legal texts.

PROFESSIONAL EXPERIENCE

Machine Learning Engineer | Slavonica Document AI d.o.o. | Osijek
March 2023–Present

- Developed a classifier for public administration documents after OCR, fine-tuning XLM-R for 23 classes. Achieved 94% accuracy and macro F1 of 0.91; the service processes approximately 15,000 documents per day.
- Fine-tuned BERTić for named entity recognition in Croatian administrative documents, including people, organisations, OIB numbers, addresses and legal references. Built an annotated dataset of 4,000 documents with guidelines and two annotators, achieving inter-annotator kappa of 0.87.
- Introduced MLflow for experiment tracking and model registry; deployed models as FastAPI services in Docker on Azure App Service, using Azure ML.
- Established weekly production drift reports covering text length, vocabulary and predicted class distribution, with alerts for PSI above 0.2. Detected an OCR change that had reduced accuracy by nine percentage points; the issue was fixed within a week.
- Mentor two student interns.

Data Scientist | Baranjska kreditna banka d.d. | Zagreb
September 2021–February 2023

- Developed customer churn prediction models using gradient boosting and LightGBM for the sales team.
- Classified customer complaints into 11 topics using TF-IDF and logistic regression, followed by fastText; supported weekly reporting to customer service management.
- Produced SQL reports using an Oracle data warehouse.

Research Project Assistant | Faculty of Applied Informatics, University of Drava
October 2020–August 2021

- Built and annotated a corpus of Croatian legal texts and wrote Python scripts for the research project.

PROJECTS

Retrieval-Augmented Question Answering Demo | Personal project | 2025

- Built a weekend RAG prototype for questions about Croatian laws using sentence embeddings, FAISS and an open-weight LLM accessed through an API.
- Evaluated the prototype through manual checks; it was not used in production.

EDUCATION

PhD in Computer Science, in progress | University of Drava | Since 2022
Research: Named entity recognition in Croatian legal texts. Expected completion: 2027.

Master of Computer Science (Mag. ing. comp.) | University of Drava | 2020

TECHNICAL SKILLS

Programming and data: Python, SQL, pandas, NumPy, Oracle SQL

Machine learning and NLP: PyTorch, Hugging Face Transformers, scikit-learn, LightGBM, fastText, spaCy, sentence-transformers, FAISS
MLOps and deployment: MLflow, FastAPI, Docker, Azure ML, Azure App Service, Azure Functions, Evidently, Git, Linux
Other tools: Label Studio, PostgreSQL

CERTIFICATION

Microsoft Certified: Azure AI Engineer Associate | 2024

PUBLICATION AND TEACHING

- Zlatarić-Novosel, P. et al. (2024). "Named entity recognition for Croatian administrative documents." Workshop on Slavic language processing, fictional proceedings.
- Guest lecture at an NLP summer school, 90 minutes | 2025

LANGUAGES

Croatian: Native | English: C1, used daily in client meetings | German: B2